Northstar Learning and Consulting

Analytics Dashboard Proposal
and Implementation

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Scope

As Northstar remains a fictional company, its business profile and proposed targets are strictly based on assumptions. Bear in mind, all the historical figures presented, come straight from the supplied workbook. The abbreviation CU represents currency units as the source file does not represent a specific currency. Recorded surplus refers to income minus the recorded expenses and does not represent audited cash flow or net profit. A completed PBIX file has been submitted separately demonstrating a practical dashboard.

Executive overview

Northstar is in need of a consistent way to track its monthly income, recorded expenses and the remaining surplus. In this proposal, we will be reviewing six different financial KPIs, a repeatable data preparation process and an interactive Microsoft Power BI report. The 2024 results shows the value in this approach. Income has decreased by 0.75% to 132,145 currency units (CU), meanwhile the recorded expenses have increased by 1.40% to 42,490 CU. The recorded surplus remains at 89,655 CU and surplus margin at 67.85%. [1] The report connects these annual results to monthly trends and provides a keen look at the individual income and expense categories.

Company overview

Northstar Learning and Consulting is a fictional business with five employees in Ottawa's professional training business advisory industry. It serves entrepreneurs and smaller organizations by means of providing workshops and consulting services online and in person. The owner remains responsible for service planning and financial decisions, meanwhile administrators prepare and maintain the monthly summaries.

The workbook provided has four income labels. They are, Consulting, Workshop, Monthly Workshop and Business income. These labels remain as they appeared in the source. The recorded expenses include internet, electricity, water, stationery, transportation, rent and other expenses. Management requires a practical means to review service income, monitor spending patterns and identifying the weaker months that require further attention. The described company profile is strictly an assumption. The workbook gives financial categories and amounts, however it does not include details regarding staffing, location, customers or other business characteristics.

Business Goals

The goals proposed are for the next comparable 12 month planning period, using the complete 2024 results as the baseline. They represent management targets instead of forecasts.

Goal	2024 baseline	Proposed target and management response
Increase annual income by 5%	132,145 CU	At least 138,752.25 CU. Review weaker income sources and investigate service demand before allocating promotional effort.
Reduce recorded expenses by 3%	42,490 CU	No more than 41,215.30 CU. Review major categories and supplier terms while protecting service delivery.
Reach a recorded surplus margin of 70%	67.85%	At least 70%. Meeting the income and expense targets together would yield 97,536.95 CU surplus and a 70.30% margin.

Purpose, Users and Supported Functions

The owner is able to use the dashboard to oversee the financial performance, plan services and review operating costs. The administrator is able to use it to reconcile monthly records and investigate unusual values within individual categories. Year and month filters make the users be able to review annual performance or focus on specific monthly comparisons. The income source chart supports service planning meanwhile the expense category charts aid in purchasing decisions and overhead management.

The monthly reviews should end with clearly documented questions, an assigned action owner and a follow up date. For instance, October's low surplus should lead to a closer review of service income and the supporting financial records. The dashboard is able to identify areas which may require investigation, however it cannot confirm causes like cancellation, attendance changes or pricing decisions without additional evidence.

KPIs and Measurement Rules

The six KPIs are created as DAX measures, using Sheet 1 as the current data source and updates monthly after the administrator validates the closed period. The category field determines if each positive amount is classified as income or as an expense.

KPI	Definition	Source / update frequency	2024 result
Total income	Sum Amount where Category = Income in the selected period.	FinanceData: Amount, Category, Date. Monthly.	132,145 CU
Recorded expenses	Sum Amount where Category = Expense in the selected period.	FinanceData: Amount, Category, Date. Monthly.	42,490 CU
Recorded surplus	Total income minus recorded expenses for the same selected period.	Derived from income and expenses. Monthly.	89,655 CU
Surplus margin	Recorded surplus ÷ total income × 100. DIVIDE handles a zero denominator.	Derived financial measures. Monthly.	67.85%
Income growth YoY	(Selected income – same prior-year period income) ÷ prior-year income × 100.	FinanceData plus DateTable[Date]. Monthly.	−0.75%
Expense growth YoY	(Selected expenses – same prior-year period expenses) ÷ prior-year expenses × 100.	FinanceData plus DateTable[Date]. Monthly.	+1.40%

Comparison logic and the corrected totals

The year over year (YoY) measures the related calendar table to move the selected date period back by one year. For instance, a full year 2024 selection is compared with 2023 while October 2024 compared to October 2023. The 2023 comparison remains blank as the dataset does not have 2022 records. The measures also return blank results where more than a year is selected, which prevents the validation page from showing a misleading combined year growth figure.

The original combined year calculation compares two years of current values with only the available prior year values. While the result remains mathematically valid, it isn't meaningful comparison. Limiting the growth calculation to one selected year corrects this context issue. All missing prior year values stay blank rather than being treated as zero.

Data Integration, Quality and Governance

Source and repeatable preparation

The supplied workbook “Monthly_Income_Expense_2023_2024.xlsx” has 264 data rows organized into four columns. They are “Amount, Date, Category and Subcategory”. The data spans over 24 months with four income entries and seven different expense entries recorded monthly. Each row represents monthly category – subcategory summary instead of an individual invoice or transaction. All the source dates use the first day of the corresponding month.

Sheet1 has been imported into Power Query and renamed “FianaceData”. Amount has been assigned a whole number data type, Date set as a date data type and the category fields formatted as text. Before cleaning was carried out, Subcategory was duplicated and renamed as “Original Subcategory” to preserve the source values. The text fields were trimmed and the spellings “Stationary” was corrected to “Stationery” in the display columns. Workshop and Monthly Workshop remain separate categories, while Business Income stays unchanged as its specific meaning was not documented.

Power Query has recorded each transformation step, which allows the extraction, transformation and loading processes to be repeated consistently. [2] Under the current process, the administrator updates the Excel source and selects Refresh in Power BI Desktop. As this is a fictional organization, the report does not currently use a live accounting connection, does not have a deployed gateway nor has a scheduled cloud refresh.

Model and calculation structure

A DAX calendar table has been created that has 731 daily rows spanning from January 1, 2023, throughout to December 31, 2024, including the 2024 leap day. The table has “Year, Month Number, Month and Year Month” fields. “DateTable(Date)” forms one side of an active one to many relationship with “FinanceData(Date). Filtering moves in a single direction from the calendar table to the financial data. Month names have been sorted by Month Number, while the trend axis gets arranged in ascending Year Month order. This calendar structure supports the time based comparisons used throughout the report. [3]

Quality controls and reconciliation

The supplied workbook was reviewed, resulting in no missing values across all four source columns and no duplicate Date-Category-Subcategory combinations. Each month has 11 entries and all amounts stand as positive numerical values. While these results support structural consistency, they do not confirm that every business expense has been recorded or that the original accounting information is completely accurate.

Before each refresh, the administrator needs to confirm that the expected columns are present, dates are valid, categories are approved, monthly keys are unique and no reporting periods are missing. Income and expense totals should then be checked with the source workbook, with all if any differences being investigated before the report get distributed. The original workbook, power Query transformation history and a dated copy of each refresh should also be kept. Any unfamiliar labels should be confirmed with the owner, rather than being changed without approval.

Access and future integration

The source workbook and PBIX files need to be stored in a restricted business folder, that is only accessible to the owner and assigned administrator. Client identifiers should not be added, unless absolutely necessary for the report's purpose.

If the business chooses to later adopt an accounting software, the manual summaries can be replaced with a controlled data collector or system export. Before the change is finalized, the organization needs to confirm the accounting basis that is used by the new source and reconcile the totals with existing records. Managed sharing and scheduled refreshes can also be added in the future. However, both would need separate configurations and a formal review of user access.

Visualization Design and Tool Recommendation

Report Layout

Area	Implemented visual	Decision supported
Header	Northstar title; Month dropdown; single-select Year dropdown. Default: all months, 2024.	Makes the selected reporting period explicit.
KPI row	Six cards for income, expenses, surplus, margin, and the two YoY rates.	Provides a concise annual or monthly financial summary.
Middle	Three-series monthly line chart with ascending Year Month on one shared CU axis.	Shows trends, seasonality candidates, and months needing investigation.
Bottom left	Income by Source: horizontal bars sorted by descending income.	Compares the four supplied income labels without inventing service profitability.
Bottom right	Recorded Expenses by Category: descending horizontal bars.	Ranks the seven recorded spending categories for review.
Validation page	Annual table containing all six measures.	Supports reconciliation and exposes unavailable YoY comparisons.

Design and Interaction

Line charts are used to show changes across the ordered monthly periods. Straight line segments were specifically used in order to avoid implying smooth or continuous activity between the monthly observations. Horizontal bar charts makes the categories easier to read and be ranked clearly. The color scheme stays consistent throughout the report, with blue representing income, orange representing expenses and teal representing surplus. Bold titles, exact bar values, legends and source note aids in providing clear readability and ensures meaning isn't only communicated through color.

The Year and Month slicers allows users to filter the report by the desired period. Interactions between the category charts are adjusted to avoid unrelated income and expense selections from producing misleading surplus calculations. The filtered October view shows that the KPI cards and category breakdowns responds well to period selections. For evidence, a screenshot is supplied in the appendix, while the PBIX file contains the complete interactive report.

Why Microsoft Power BI fits this case

Ease of Use

Power BI imports Excel data, Power Query shows visible transformation steps and drag and drop visuals makes Power BI practical for a small team. Also, DAX supports reusable calculations, however the user must understand filter context and take responsibility for maintaining the model.

Integration

The current Excel connection meets the course's assignment requirement and also provides a clear path for connection of an accounting software or database sources in the future.

Cost

Using Power BI Desktop avoids the need for cloud deployment for this assignment. However, organizational sharing costs would be heavily dependent on licensing and selected capacity. [4]

Scalability

A separate calendar table and reusable measures provide solid foundation as the organization builds on monthly history. Additional financial sources can be integrated once for common rules for categories and dates have been established. A production rollout still needs controlled refreshes, appropriate permissions, performance testing and frequent reconciliation.

Excel would have been able to support the basic static financial review, however Power BI is a better choice for demonstrating the interactive filtering and reusable analytical model required for this project.

Discoveries and Practical Follow-Up

Discoveries

Period	Income (CU)	Expenses (CU)	Surplus (CU)	Margin
2023	133,148	41,903	91,245	68.53%
2024	132,145	42,490	89,655	67.85%
Both years	265,293	84,393	180,900	68.19%

Practical Follow-Up

Annual Performance

In 2024, the income had decreased by 1,003 CU while the recorded expenses went up by 587 CU. Together, these reduced the recorded surplus by 1,590 CU, or 1.74%. Due to this, the owner needs to consider income growth and cost control together rather than evaluating performance based on only income.

Monthly Exception

October 2024 has recorded an income of 5,505 CU, expenses of 3,882 CU and surplus of 1,623 CU. The monthly margin was recorded at 29.48%. This is lower than the annual margin which is recorded at 67.85%. Income also went down by 6.46% compared to October 2023. The related service records needs to be reviewed before the results are linked to changes in demand, pricing or cancellations.

Category Review

Consulting was the largest source of income in 2024, which contributed 36,757 CU. Internet had the largest expense which was 7,481 CU. These findings help to identify areas that need further review, however they do not prove that Consulting is the most profitable service or that the internet costs are excessive. Before any decision is made, the owner needs to examine the relevant contracts and operational requirements.

Appendix A. Full-Year Financial Overview

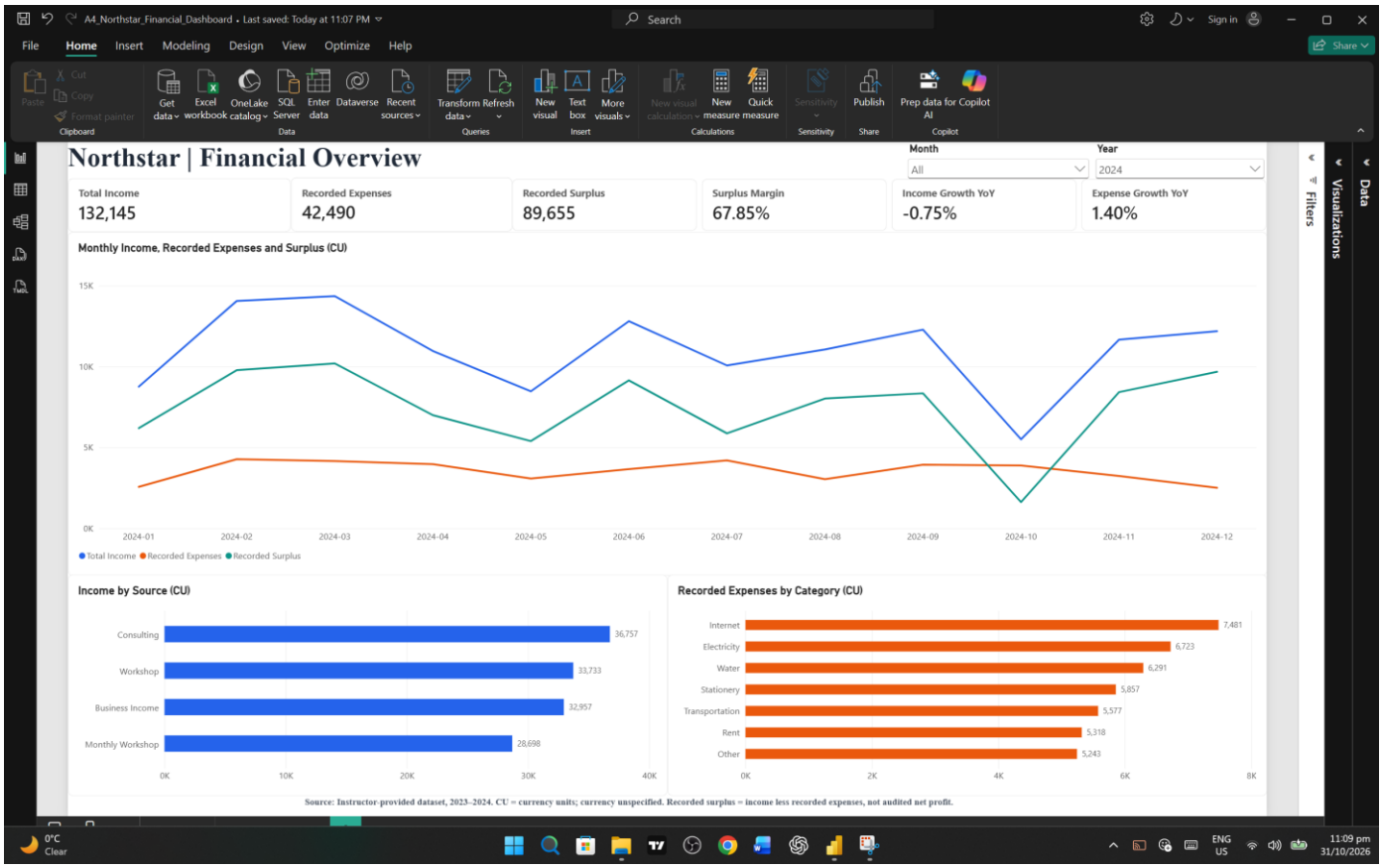


Figure 1: Full Year Financial Overview.

Appendix B. October Filter Demonstration

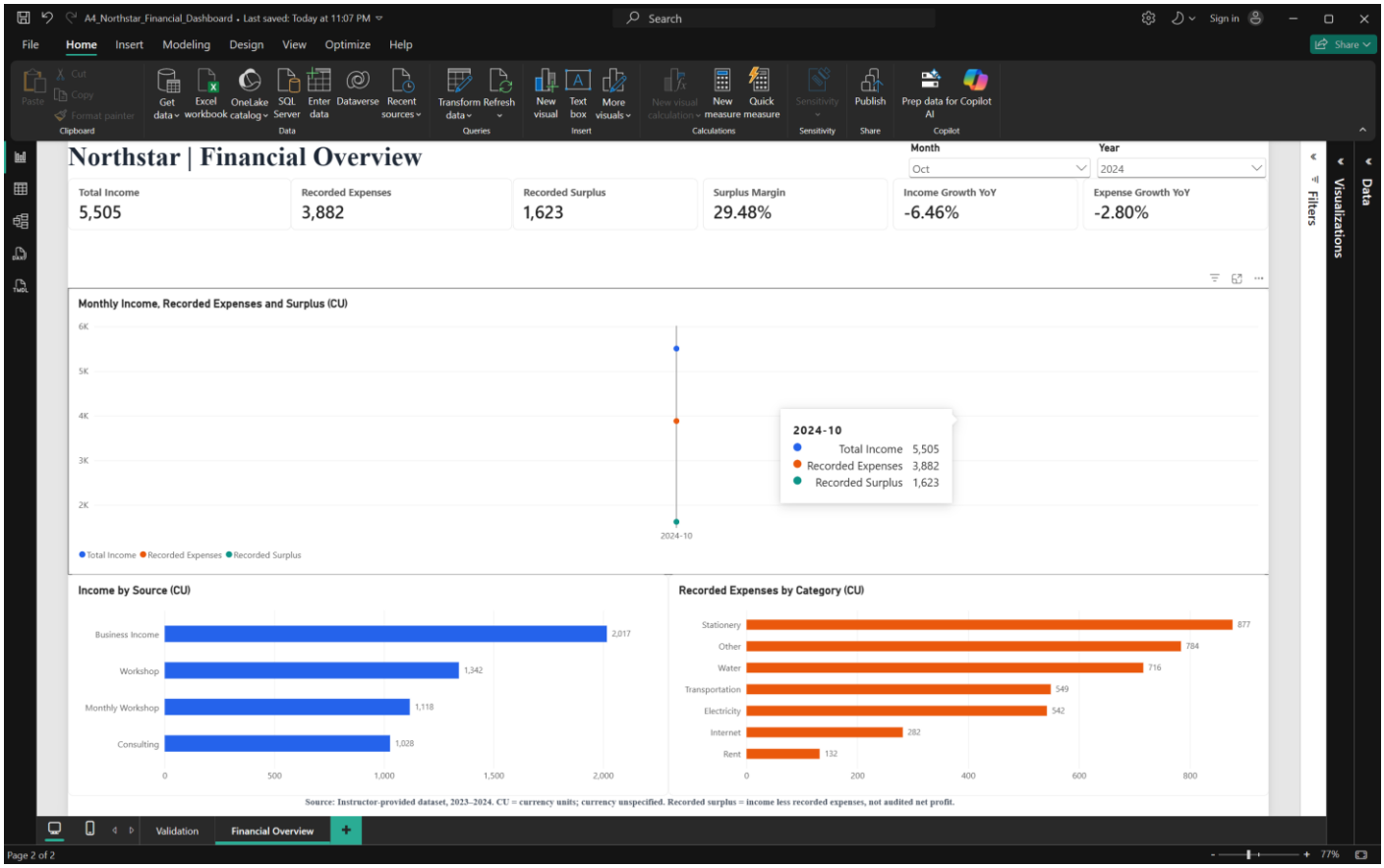


Figure 2: October Filter Demonstration.

Practical references

[1] Instructor-provided course dataset. (2026). Monthly_Income_Expense_2023_2024.xlsx, Sheet1. Historical figures and checks in this report were calculated from this workbook.

[2] Microsoft. (n.d.). What is Power Query? Microsoft Learn. https://learn.microsoft.com/en-us/power-query/power-query-what-is-power-query

[3] Microsoft. (n.d.). Design guidance for date tables in Power BI Desktop. Microsoft Learn. https://learn.microsoft.com/en-us/power-bi/guidance/model-date-tables

[4] Microsoft. (n.d.). Power BI service features by license type. Microsoft Learn. https://learn.microsoft.com/en-us/power-bi/fundamentals/service-features-license-type